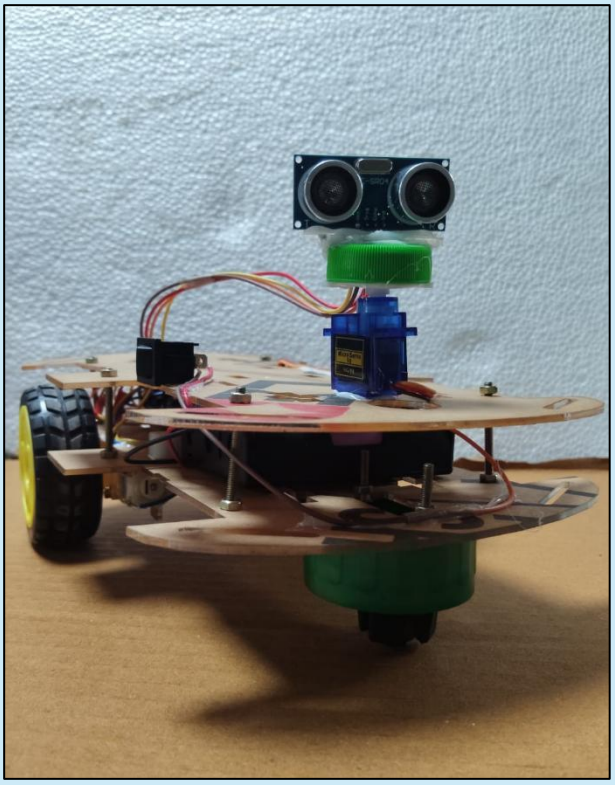
# Obstacle Avoidance Car

-By Class 12th Science



## Working

- **Power:** A 9V battery runs the Arduino and the motors.
- **Ping:** The sensor sends a silent 40 kHz pulse.
- **Echo:** The pulse bounces back, and the Arduino times it: distance = (time × speed of sound) ÷ 2.
- **Clear (over 15 cm):** Both motors drive forward.
- **Obstacle (under 10 cm):** Stop, scan right and left, reverse, turn left, then drive on and repeat.

## Think about them:

1. On a hot day, will the car misjudge a 50 cm obstacle as nearer or farther?
2. The car reverses with no rear sensor. What could go wrong?
3. Why is “turn for 500 ms” an unreliable way to turn a fixed angle?
4. What happens if you add two more sensors and fire them together?
5. Why does the code check distance > 0 as well as < 10?
6. Why might the car miss an obstacle only 1 cm away?